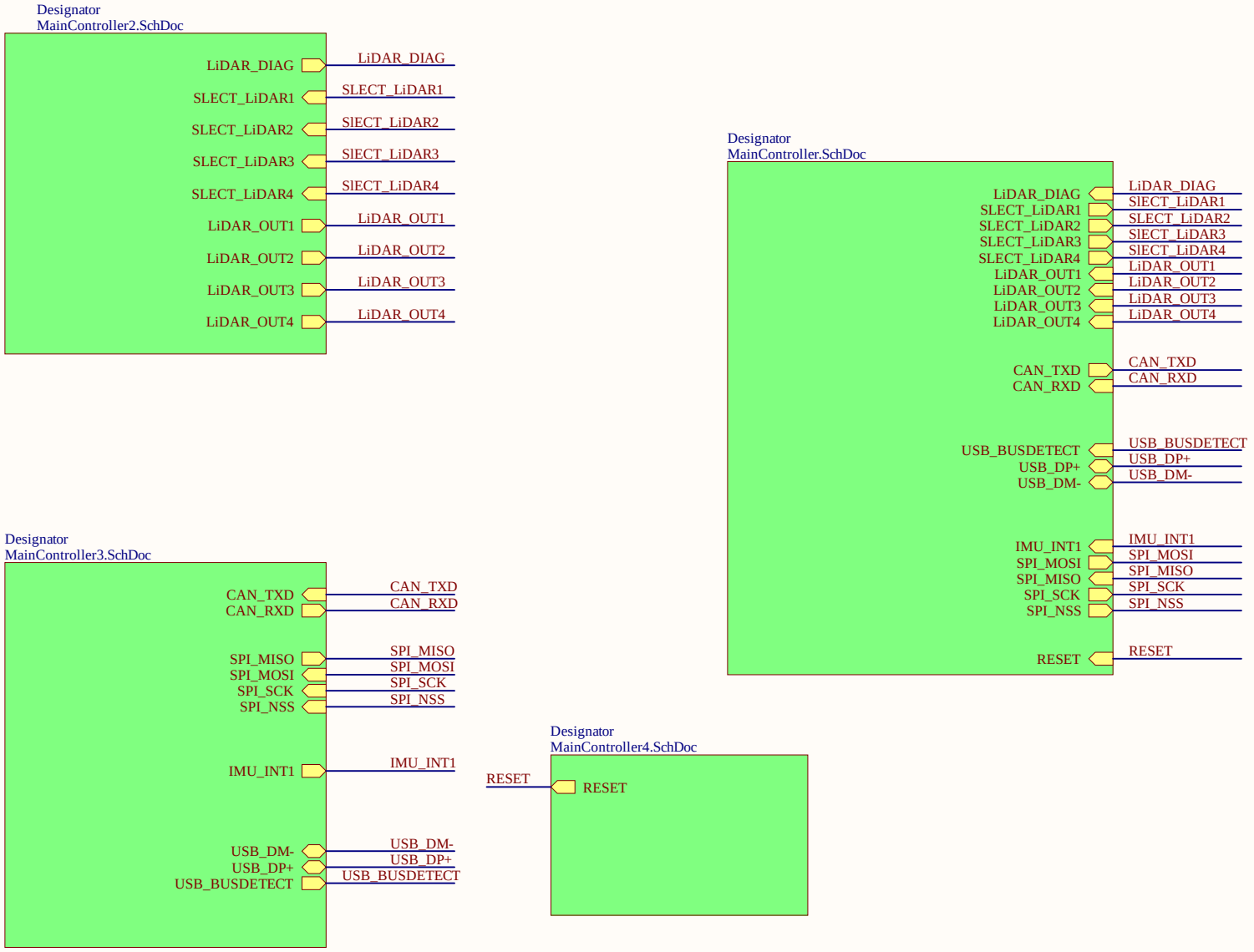
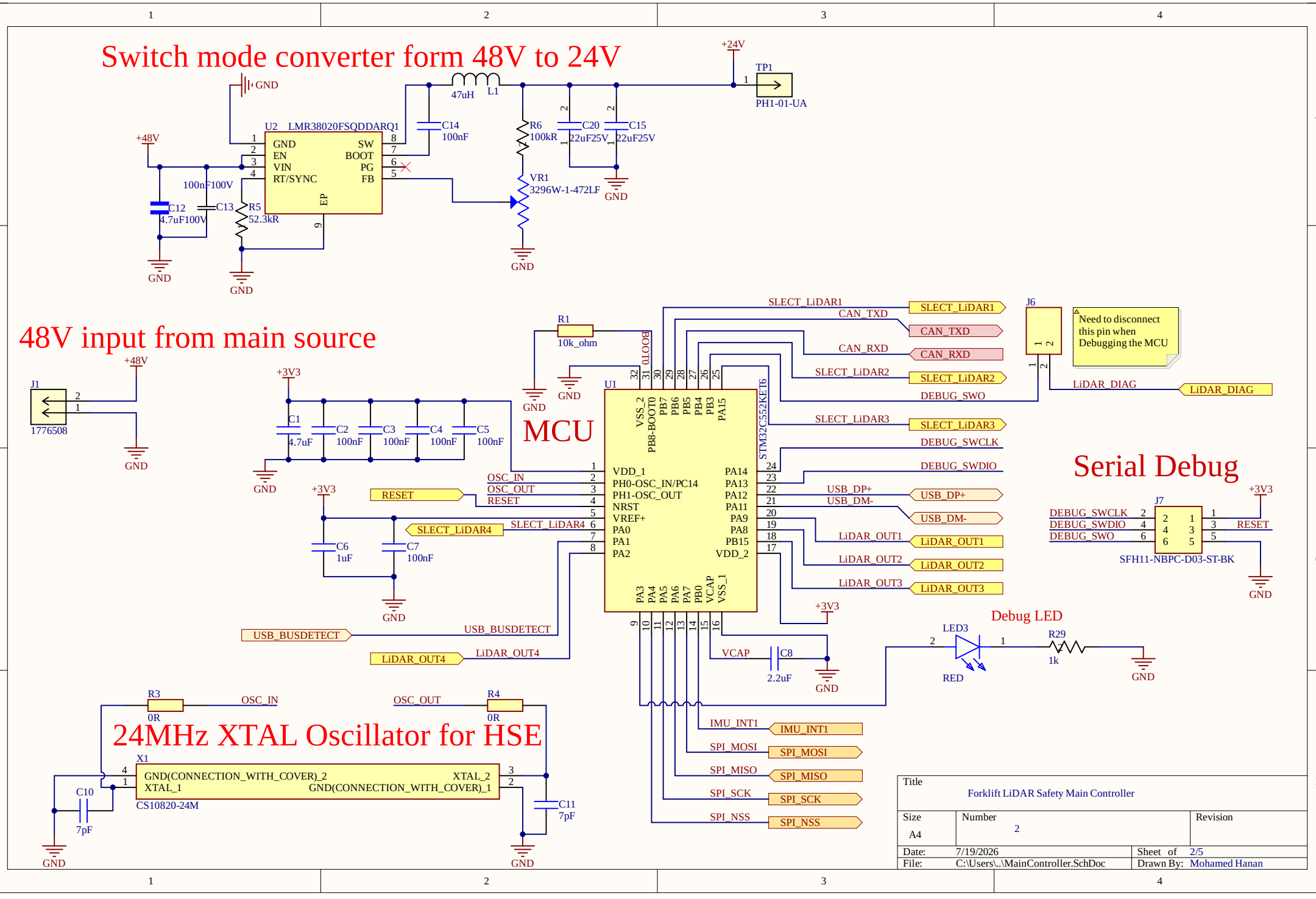
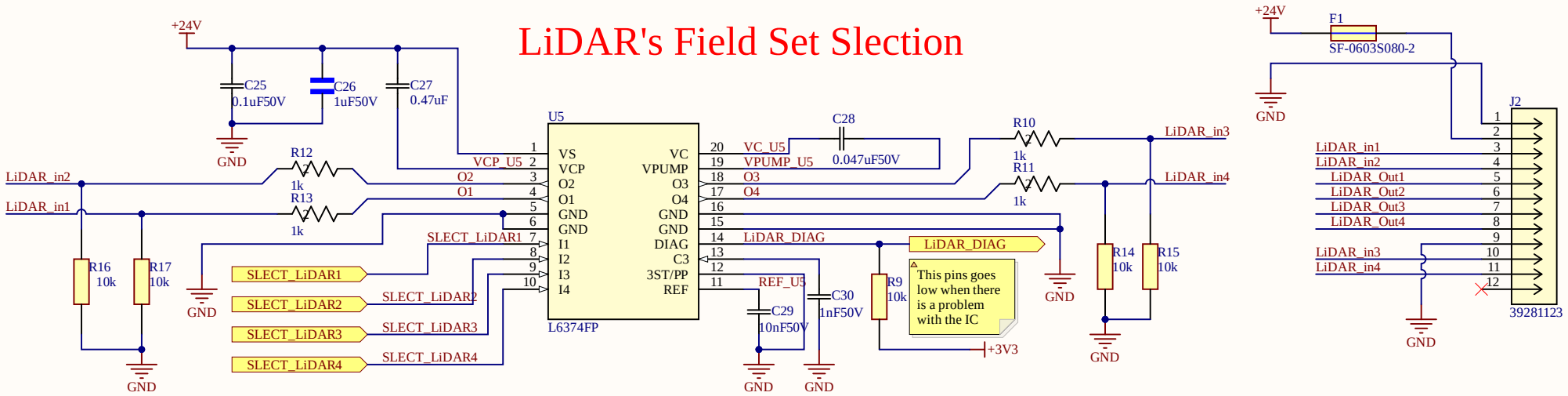
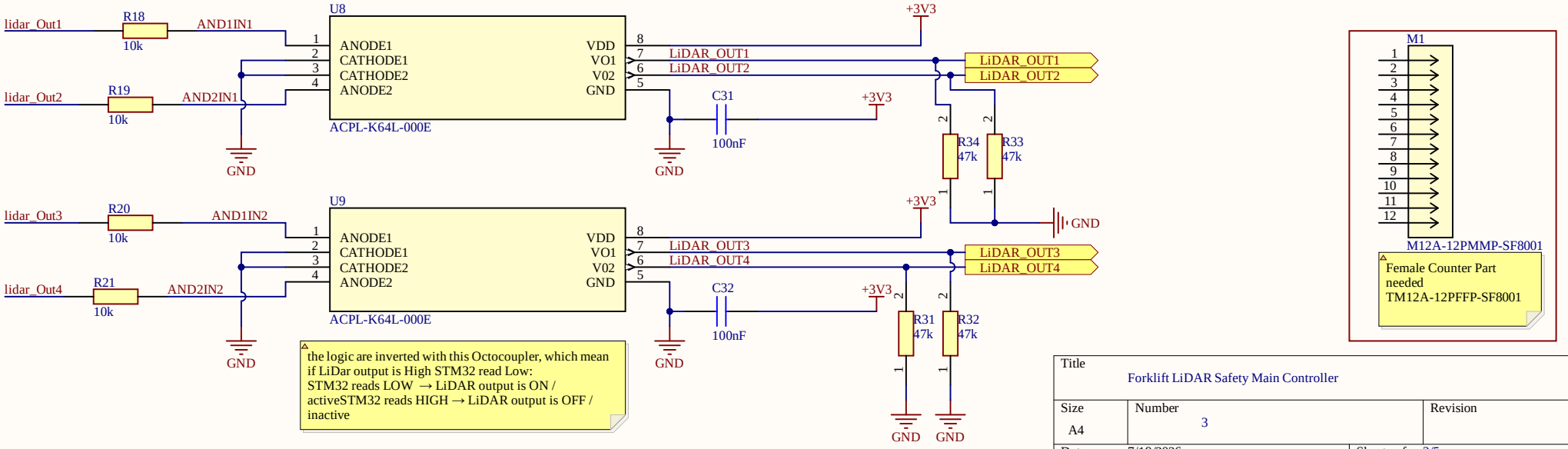

Designator
MainController2.SchDoc
LiDAR_DIAG
SLECT_LiDAR1
SLECT_LiDAR2
SLECT_LiDAR3
SLECT_LiDAR4
LiDAR_OUT1
LiDAR_OUT2
LiDAR_OUT3
LiDAR_OUT4
Designator
MainController.SchDoc
CAN_TXD
CAN_RXD
USB_BUSDETECT
USB_DP+
USB_DM-
IMU_INT1
SPI_MOSI
SPI_MISO
SPI_SCK
SPI_NSS
RESET
Designator
MainController3.SchDoc
Designator
MainController4.SchDoc
Title
Forklift LiDAR Safety Main Controller
Size
A4
Number
1
Revision
Date:
7/19/2026
Sheet of
1/5
File:
C:\Users\..\ConnectionMain.SchDoc
Drawn By:
Mohamed Hanan


Switch mode converter form 48V to 24V
+24V
TP1
PH1-01-UA
GND
47uH
L1
U2 LMR38020FSQDDARQ1
GND
EN
VIN
RT/SYNC
SW
BOOT
PG
FB
EP
+48V
C14
100nF
R6
100kR
C20
22uF25V
C15
22uF25V
VR1
3296W-1-472LF
100nF100V
C12
4.7uF100V
C13
R5
52.3kR
48V input from main source
J1
1776508
+3V3
C1
4.7uF
C2
100nF
C3
100nF
C4
100nF
C5
100nF
R1
10k_ohm
MCU
U1
STM32C552KET6
VDD_1
PH0-OSC_IN/PC14
PH1-OSC_OUT
NRST
VREF+
PA0
PA1
PA2
PA14
PA13
PA12
PA11
PA9
PA8
PB15
VDD_2
OSC_IN
OSC_OUT
RESET
SLECT_LiDAR4
C6
1uF
C7
100nF
USB_BUSDETECT
LiDAR_OUT4
VCAP
C8
2.2uF
SLECT_LiDAR1
CAN_TXD
CAN_RXD
SLECT_LiDAR2
DEBUG_SWO
SLECT_LiDAR3
DEBUG_SWCLK
DEBUG_SWDIO
USB_DP+
USB_DM-
LiDAR_OUT1
LiDAR_OUT2
LiDAR_OUT3
IMU_INT1
SPI_MOSI
SPI_MISO
SPI_SCK
SPI_NSS
J6
Need to disconnect this pin when Debugging the MCU
LiDAR_DIAG
Serial Debug
J7
SFH11-NBPC-D03-ST-BK
Debug LED
LED3
RED
R29
1k
24MHz XTAL Oscillator for HSE
R3
0R
R4
0R
X1
CS10820-24M
GND(CONNECTION_WITH_COVER)_2
XTAL_1
XTAL_2
GND(CONNECTION_WITH_COVER)_1
C10
7pF
C11
7pF
Title
Forklift LiDAR Safety Main Controller
Size
A4
Number
2
Revision
Date:
7/19/2026
Sheet of
2/5
File:
C:\Users\..\MainController.SchDoc
Drawn By:
Mohamed Hanan

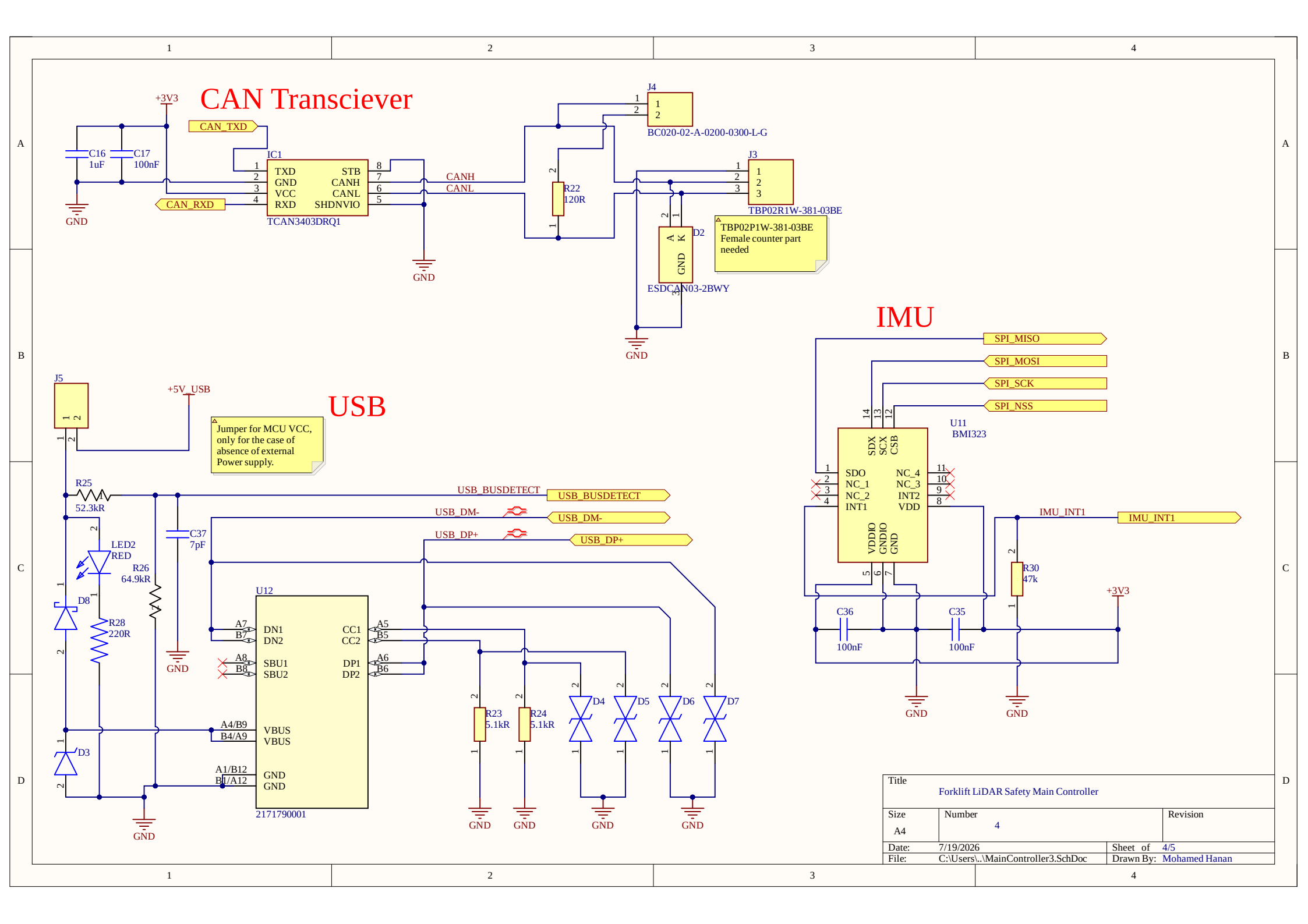
LiDAR's Field Set Slection



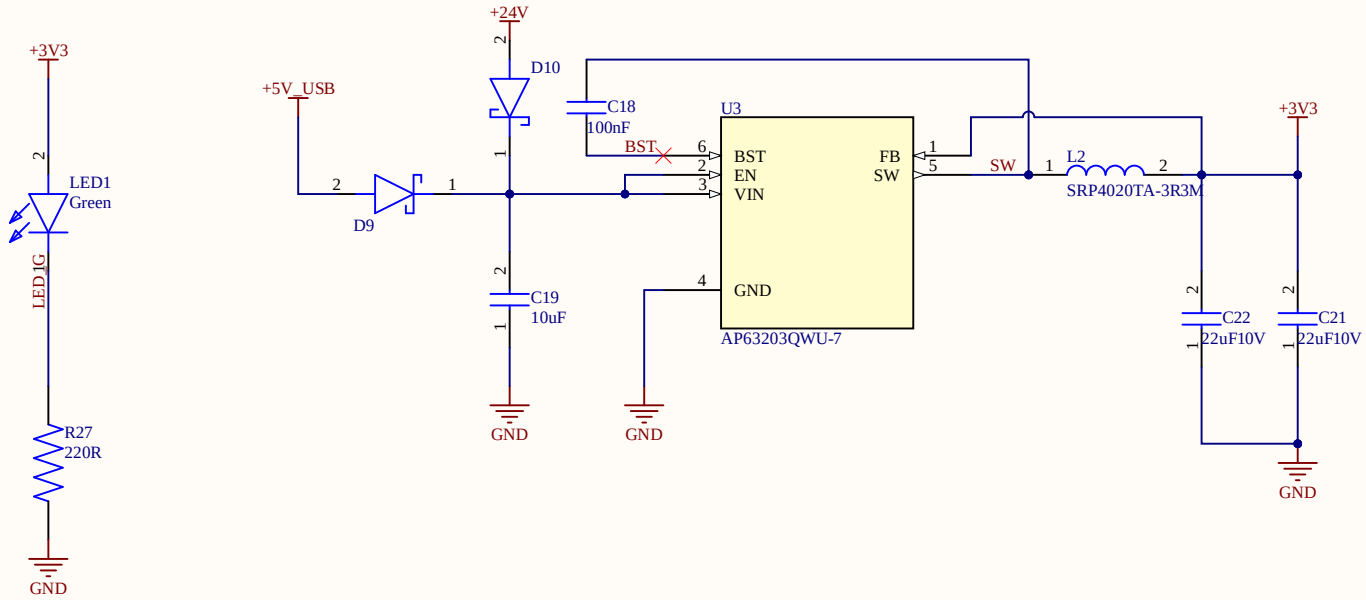
LiDAR's Logic Output



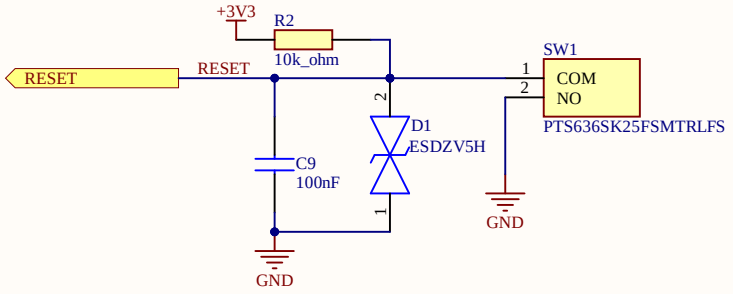
Title	Forklift LiDAR Safety Main Controller	
Size: A4	Number: 3	Revision:
Date: 7/19/2026		Sheet of 3/5
File: C:\Users\..\MainController2.SchDoc		Drawn By: Mohamed Hanan


CAN Transciever
+3V3
CAN_TXD
CAN_RXD
C16
1uF
C17
100nF
IC1
TXD
GND
VCC
RXD
STB
CANH
CANL
SHDNVIO
TCAN3403DRQ1
GND
CANH
CANL
J4
BC020-02-A-0200-0300-L-G
R22
120R
J3
TBP02R1W-381-03BE
TBP02P1W-381-03BE
Female counter part
needed
D2
A
K
GND
ESDCAN03-2BWY
GND
IMU
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_NSS
U11
BMI323
SDX
SCX
CSB
SDO
NC_1
NC_2
INT1
NC_4
NC_3
INT2
VDD
VDDIO
GNDIO
GND
IMU_INT1
R30
47k
+3V3
C36
100nF
C35
100nF
GND
GND
USB
J5
+5V_USB
Jumper for MCU VCC,
only for the case of
absence of external
Power supply.
R25
52.3kR
USB_BUSDETECT
USB_DM-
USB_DP+
C37
7pF
LED2
RED
R26
64.9kR
D8
R28
220R
D3
U12
DN1
DN2
SBU1
SBU2
VBUS
VBUS
GND
GND
CC1
CC2
DP1
DP2
A7
B7
A8
B8
A5
B5
A6
B6
A4/B9
B4/A9
A1/B12
B1/A12
2171790001
R23
5.1kR
R24
5.1kR
D4
D5
D6
D7
GND
GND
GND
GND
GND
Title
Forklift LiDAR Safety Main Controller
Size
A4
Number
4
Revision
Date:
7/19/2026
Sheet of
4/5
File:
C:\Users\..\MainController3.SchDoc
Drawn By:
Mohamed Hanan

Switch mode Regulator 24V to 3.3V



Switch 4 MCU RESET PIN



Title: Forklift LiDAR Safety Main Controller		
Size: A4	Number: 5	Revision:
Date: 7/19/2026		Sheet of 5/5
File: C:\Users\..\MainController4.SchDoc		Drawn By: Mohamed Hanan

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT	C1	CAPC1608X90N	CL10A475KP8NNNC	1
100nF	Samsung Electro-Mechanics CL05B104KO5NNNC 100nF Multilayer Ceramic Capacitor MLCC 16Vdc +/-10% Tolerance SMD	C2, C3, C4, C5, C7, C9, C14, C17, C18, C31, C32, C35, C36	CAPC1005X55N	CL05B104KO5NNNC	13
1uF	Samsung Electro-Mechanics 0402 (1005M) 1uF Multilayer Ceramic Capacitor MLCC 16Vdc +/-10% SMD CL05A105KO5NNNC	C6, C16	CAPC1005X55N	CL05A105KO5NNNC	2
2.2uF	Samsung Electro-Mechanics 0603 CL 2.2uF Ceramic Multilayer Capacitor, 10 V, +85C, X5R Dielectric, +/-10% SMD	C8	CAPC1608X90N	CL10A225KP8NNNC	1
7pF	CAPCER 7PF 50V C0G/NP0 0603	C10, C11, C37	CAPC1608X90N	CL10C070CB8NNNC	3
4.7uF/100V	SMD capacitor X7S(EIA) with capacitance 4.7uF Tol.20% Rated voltage 100Vdc 125C	C12	CAPC3216X190N	GRM31CC72A475ME1 1L	1
100nF/100V	Description	C13	CAP_0603_VAL	0603B104K101CT	1
22uF/25V	Capacitor	C15, C20	CAPC3216X190N	TMK316BBJ226ML-T	2
10uF	Capacitor	C19	CAPC3216X180N	CL31A106MAHNNNE	1
22uF/10V	Capacitor	C21, C22	GRM18x	GRM188R61A226ME1 5D	2
0.1uF/50V	Description	C25	CAP_GCM1555_0402_MUR	GCM155R71H104KE0 2J	1
1uF/50V	SMD capacitor X5R(EIA) with capacitance 1.0uF Tol.20% Rated voltage 50Vdc 85C	C26	CAPC1005X60N	GRM155R61H105ME0 5D	1
0.47uF	Description	C27	CAP_GRM155_MUR	GRM155R61H474KE1 1D	1
0.047uF/50V	Description	C28	CAP_CL10_SAM	CL10B473KB8NNNC	1
10nF/50V	SMD Comm.X7R, Ceramic, 0.01 uF, 10%, 50 VDC, X7R, SMD, MLCC, Temperature Stable, Class II, 0603	C29	CAPC1759S_87N_KEM	C0603C103K5RACTU	1
1nF/50V	Description	C30	CAP_GRM_155_MUR	GRM1555C1H102JA01 D	1
ESD2V5H	Ultra low clamping single line bidirectional ESD protection	D1	ESD2V5H1BL2	ESD2V5H-1BL2	1
ESDCAN03-2BWY	TVS Diode (Uni-directional)	D2	SOT65P210X110-3N	ESDCAN03-2BWY	1
ESDA25P35-1U1M	TVS Diode (Uni-directional)	D3	ESDA25P351U1M	ESDA25P35-1U1M	1
ESD2V5H-1BL2	Ultra low clamping single line bidirectional ESD protection	D4, D5, D6, D7	ESD2V5H1BL2	ESD2V5H-1BL2	4
VS-30BQ015-M3_9AT	Schottky Diode	D8, D9	DIOM7959X262N	VS-30BQ015-M3_9AT	2
PSOD01LA40/TR	Schottky Diode	D10	SODFL4725X105N	PSOD01LA40/TR	1
SF-0603SD80-2	BOURNS - SF-0603SD80-2 - FUSE, 0.8A, 32V, 0603	F1	FUSC1608X65N	SF-0603SD80-2	1
TCAN3403DRQ1	Integrated Circuit	IC1	SOIC127P600X175-8N - duplicate	TCAN3403DRQ1	1
1776508	PCB headers, nominal current: 12 A, rated voltage (III/2): 320 V, number of positions: 2, pitch: 5.08 mm, color: green, contact surface: Tin, mounting: Wave soldering, Gold-plated contact surface / color: Green RAL 6021	J1	CONN_1776508	1776508	1
39281123	Mini-Fit Jr. Vertical Header, 4.20mm Pitch, Dual Row, 12 Circuits, without Snap-in Plastic Peg PCB Lock, Tin, Natural	J2	CONN_SD-5566-002_12_MOL	39281123	1
TBP02R1W-381-038E	Connector	J3	RHDRR3V80P2X381_1X2_21826X20X790P	TBP02R1W-381-038E	1
BC020-02-A-0200-0300-LG	Connector	JA, JB, JE	BC0200240200300LG	BC020-02-A-0200-0300-LG	3
SFH11-NBPC-D03-SF-BK	Connector	JF	SFH11NBPCD03STBK	SFH11-NBPC-D03-SF-BK	1
47uH	Description	L1	IND_PA4343/PA4343_FP_FLL	PA4343.473NLT	1
SRP4020TA-3R3M	Inductor	L2	SRP4020TA5R6M	SRP4020TA-3R3M	1
Green	LED	LED1	LEDC1608X65N	LTST-C190GKT	1
RED	LED	LED2, LED3	LEDC1608X65N	LTST-C191KRKT	2
10k_ohm	YAGEO (PHYCOMP) RC0603 F, 10k OHM, 0.1W	R1, R2	RESC1608X55N	RC0603FR0710KL	2
0R	Res Thick Film 0402 0 Ohm Epoxy Pad SMD TR	R3, R4	RESC1005X40N	RC0402JR070RL	2
52.3kR	SMD RESS2.3K.1/16W.1%.0402	R5, R25	RES_RC0402_YAG	RC0402FR-0752K3L	2
100kR	011 7 CRCW0603 100 100K 1% ET1 e3	R6	RES_CRCW_0603_VIS	CRCW0603100KFKEA	1
10k	YAGEO (PHYCOMP) RC0603 F	R9, R14, R15, R16, R17, R18, R19, R20, R21	RESC1608X55N	RC0603FR0710KL	9
1k	011 7 CRCW0603 100 1K 1% ET1 e3	R10, R11, R12, R13, R29	RES_CRCW_0603	CRCW06031K00FKEA	5
120R	Resistor	R22	RCS1206 E3	RCS120612R0FKEA	1
5.1kR	Resistor	R23, R24	RESC1608X55N	RC0603FR075K1L	2
643kR	Description	R26	RC0603N_YAG	RC0603FR-07643KL	1
220R	Fixed Resistor CRCG 0603 220R 5%	R27, R28	RESC1608X55N	CRCG0603J220R	2
47k	Resistor	R30, R31, R32, R33, R34	RESC1005X40N	RC0402FR-0747KL	5
PTS636 Series SPST-NO Off-Mom 250 gf Top Actuated SMT Tactile Switch	PTS636 Series SPST-NO Off-Mom 250 gf Top Actuated SMT Tactile Switch	SW1	PTS636 3 K25 F SMTR LFS	PTS636 3 K25 F SMTR LFS	1
PH1-01-UA	Description	TP1	CONN_PH1-01-UA_ADM	PH1-01-UA	1
STM32C552KET6	Arm® Cortex®-M33 32-bit MCU with FPU, 144 MHz, SB3 CoreMark®, up to 512Kbyte dual bank flash memory, 128-Kbyte RAM	U1	QFP80P9004X9.00X160-32N	STM32C552KET6	1
LMR38020FSQDDARQ1	Switching Voltage Regulators Automotive SIMPLE SWITCHER power converter 4.2-V to 80-V, 2-A, synchronous buck converter	U2	SOIC127P600X170-9N	LMR38020FSQDDARQ1	1
AP63203QWU-7	Description	U3	SOT26_DIO	AP63203QWU-7	1
L6374FP	INDUSTRIAL QUAD LINE DRIVER	U5	SO-20_STM	L6374FP	1
ACPL-K64L-000E	Description	U8, U9	STRETCHED_SO-8_AVA	ACPL-K64L-000E	2
BM323	Integrated Circuit	U11	BM323	BM323	1
2171790001	Universal Serial Bus (USB) Shielded I/O Receptacle, Type-C, Right-Angle, Top Mount, Surface Mount, Single Row, 16 Circuits, Gold (Au) Plating	U12	2171790001_MOL	2171790001	1
3296W-1-472LF	3296W top adj cermet trimmer 4k7 10% PCB,25 Turns,Part Rtg:0.5W) Bourns 3296W-1-472LF	VR1	TRIM_3296W-1-472LF	3296W-1-472LF	1
CS10820-24M	24 MHz +/- 15ppm Crystal 6pF 80 Ohms 4-SMD, No Lead , -40C ~ 125C	X1	CS10820024M	CS10820-24M	1